Ethical Hacking Lab 3

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Port Scanning & SSH Tunneling

Q) On a Centos VM perform the following.

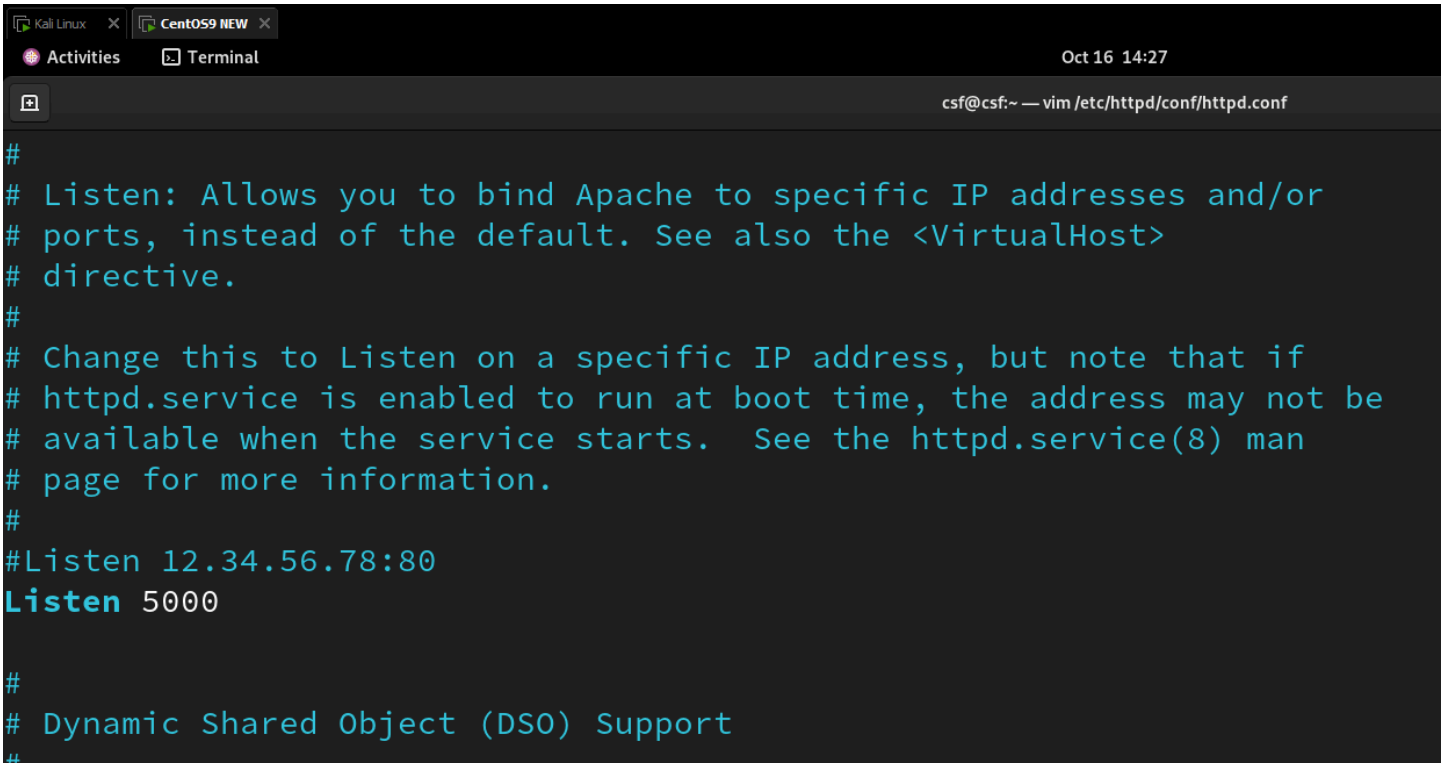
- Install httpd web server. Configure it to run on port 5000. Start the service. Enable the service to start automatically.
- Install nginx web server. It will run on port 80. Start and enable the nginx service.
- Open ports 80 and 5000 in the CentOS firewall.
- Open a Terminal and start python http server to run on port 8500. Do not open this port in the firewall.

Now from the Kali linux

- Perform nmap scan on CentOS VM IP address and display the ports shown by nmap.
- Use metasploit to perform the port scanning on CentOS VM using nmap and auxiliary port scanner and display the result.
- From Kali linux create a SSH tunnel and access the internal python website on port 10000 on Kali Linux.

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1. Edited **httpd** config to listen on port 5000
   sudo vim /etc/httpd/conf/httpd.conf



Setting SELinux on permissive mode

```
[csf@csf ~]$ sudo getenforce
Enforcing
[csf@csf ~]$ sudo setenforce 0
[csf@csf ~]$ sudo getenforce
Permissive
```

Checking status of httpd

```
[csf@csf ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: active (running) since Fri 2026-10-16 14:33:35 IST; 14s ago
       Docs: man:httpd.service(8)
   Main PID: 33468 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 22780)
     Memory: 14.6M (peak: 15.5M)
        CPU: 98ms
     CGroup: /system.slice/httpd.service
             ├─33468 /usr/sbin/httpd -DFOREGROUND
             ├─33480 /usr/sbin/httpd -DFOREGROUND
             ├─33481 /usr/sbin/httpd -DFOREGROUND
             ├─33482 /usr/sbin/httpd -DFOREGROUND
             └─33483 /usr/sbin/httpd -DFOREGROUND

Oct 16 14:33:10 csf systemd[1]: Starting The Apache HTTP Server...
Oct 16 14:33:22 csf httpd[33468]: AH00558: httpd: Could not reliably determine the server's fully qualified do>
Oct 16 14:33:35 csf httpd[33468]: Server configured, listening on: port 5000
```

2. Installing ngnix

```
[csf@csf ~]$ sudo dnf install nginx
Last metadata expiration check: 0:26:01 ago on Friday 16 October 2026 02:09:35 PM.
Package nginx-2:1.20.1-32.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[csf@csf ~]$
```

Enabling and starting ngnix

```
[csf@csf ~]$ sudo systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/syst
ce.
[csf@csf ~]$ sudo systemctl start nginx
[csf@csf ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/nginx.service.d
             └─php-fpm.conf
     Active: active (running) since Fri 2026-10-16 14:37:28 IST; 6s ago
    Process: 33775 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 33776 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 33777 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 33779 (nginx)
      Tasks: 5 (limit: 22780)
     Memory: 5.3M (peak: 5.5M)
        CPU: 32ms
     CGroup: /system.slice/nginx.service
             ├─33779 "nginx: master process /usr/sbin/nginx"
             ├─33780 "nginx: worker process"
             ├─33781 "nginx: worker process"
             ├─33782 "nginx: worker process"
             └─33783 "nginx: worker process"
```

3. Opening port numbers 80 and 5000 on firewall

```
[csf@csf ~]$ sudo firewall-cmd --permanent --add-port=80/tcp
Warning: ALREADY_ENABLED: 80:tcp
success
[csf@csf ~]$ sudo firewall-cmd --add-port=80/tcp
success
[csf@csf ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 22/tcp 80/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[csf@csf ~]$
```

```
[csf@csf ~]$ sudo firewall-cmd --permanent --add-port=5000/tcp
success
[csf@csf ~]$ sudo firewall-cmd --add-port=5000/tcp
success
[csf@csf ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 22/tcp 80/tcp 5000/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[csf@csf ~]$
```

Created web directory and inside it created index.html file

```
[csf@csf web]$ cat index.html
<h1>Hello</h1>
[csf@csf web]$
```

Running HTTP Python server on port 8500

```
[csf@csf web]$ ls
index.html
[csf@csf web]$ python3 -m http.server 8500
Serving HTTP on 0.0.0.0 port 8500 (http://0.0.0.0:8500/) ...
```

Checking CentOS IP address

```
[csf@csf web]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qle
    link/ether 00:0c:29:c0:97:6a brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.35.20/24 brd 192.168.35.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet 192.168.200.132/24 brd 192.168.200.255 scope global dynamic noprefixroute ens160
       valid_lft 1249sec preferred_lft 1249sec
    inet6 fe80::20c:29ff:fec0:976a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[csf@csf web]$
```

==========

On kali

Performed nmap scan on CentOS VM IP address

```
┌──(kali㉿kali)-[~]
└─$ nmap -Pn -sS -sV 192.168.200.132 -p 22,80,443,5000,8500
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-16 05:26 -0400
Nmap scan report for 192.168.200.132
Host is up (0.0017s latency).

PORT     STATE    SERVICE VERSION
22/tcp   open     ssh     OpenSSH 9.9 (protocol 2.0)
80/tcp   open     http    nginx 1.20.1
443/tcp  filtered https
5000/tcp open     http    Apache httpd 2.4.62 ((CentOS Stream))
8500/tcp filtered fmtp
MAC Address: 00:0C:29:C0:97:6A (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.84 seconds
```

Using Metasploit, selected an auxiliary port scanner

```
kali@kali: ~   kali@kali: ~

msf > search portscan

Matching Modules
================

   #  Full Name                                          Disclosure Date  Rank    Check
   -  ---------                                          ---------------  ----    -----
   0  auxiliary/scanner/portscan/ftpbounce               .                normal  No
   1  auxiliary/scanner/natpmp/natpmp_portscan           .                normal  No
   2  auxiliary/scanner/sap/sap_router_portscanner       .                normal  No
   3  auxiliary/scanner/portscan/xmas                    .                normal  No
   4  auxiliary/scanner/portscan/ack                     .                normal  No
   5  auxiliary/scanner/portscan/tcp                     .                normal  No
   6  auxiliary/scanner/portscan/syn                     .                normal  No
   7  auxiliary/scanner/http/wordpress_pingback_access   .                normal  No


Interact with a module by name or index. For example info 7, use 7 or use auxiliary/sca

msf > use 6
msf auxiliary(scanner/portscan/syn) > options
```

Checked the options and set RHOSTS and Threads as 4

```
msf auxiliary(scanner/portscan/syn) > options

Module options (auxiliary/scanner/portscan/syn):

   Name       Current Setting  Required  Description
   ----       ---------------  --------  -----------
   BATCHSIZE  256              yes       The number of hosts to sca
   DELAY      0                yes       The delay between connecti
   INTERFACE                   no        The name of the interface
   JITTER     0                yes       The delay jitter factor (m
   PORTS      1-10000          yes       Ports to scan (e.g. 22-25,
   RHOSTS                      yes       The target host(s), see ht
   SNAPLEN    65535            yes       The number of bytes to cap
   THREADS    1                yes       The number of concurrent t
   TIMEOUT    500              yes       The reply read timeout in


View the full module info with the info, or info -d command.

msf auxiliary(scanner/portscan/syn) > set RHOSTS 192.168.200.132
RHOSTS ⇒ 192.168.200.132
```

Performed the port scanning using auxiliary port scanner

```
msf auxiliary(scanner/portscan/syn) > run
/usr/share/metasploit-framework/lib/msf/core/exploit/capture.
UB::Pcap
[+]  TCP OPEN 192.168.200.132:22
[+]  TCP OPEN 192.168.200.132:80
[+]  TCP OPEN 192.168.200.132:5000
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/portscan/syn) >
```

Using db_nmap to scan the target.

```
msf auxiliary(scanner/portscan/syn) > db_nmap -sS -sV 192.168.200.132
[*] Nmap: Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-16 05:30 -0400
[*] Nmap: Nmap scan report for 192.168.200.132
[*] Nmap: Host is up (0.0010s latency).
[*] Nmap: Not shown: 985 filtered tcp ports (no-response), 11 filtered tcp ports (admi
[*] Nmap: PORT     STATE  SERVICE    VERSION
[*] Nmap: 22/tcp   open   ssh        OpenSSH 9.9 (protocol 2.0)
[*] Nmap: 80/tcp   open   http       nginx 1.20.1
[*] Nmap: 5000/tcp open   http       Apache httpd 2.4.62 ((CentOS Stream))
[*] Nmap: 9090/tcp closed zeus-admin
[*] Nmap: MAC Address: 00:0C:29:C0:97:6A (VMware)
[*] Nmap: Service detection performed. Please report any incorrect results at https://
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 16.73 seconds
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.35/lib/recog/fi
ng: nested repeat operator '+' and '?' was replaced with '*' in regular expression
msf auxiliary(scanner/portscan/syn) >
```

Using SSH tunneling, tunneled victim's localhost 8500 to attacker's localhost 10000

```
kali@kali: ~   kali@kali: ~

┌──(kali㉿kali)-[~]
└─$ ssh -N -L localhost:10000:localhost:8500 csf@192.168.200.132
The authenticity of host '192.168.200.132 (192.168.200.132)' can't be established.
ED25519 key fingerprint is: SHA256:QTCNCo2F7dqiaCl/pGtHjysTM1h3EqD8PGOmFCpLUoI
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.200.132' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
csf@192.168.200.132's password:
```

Visited localhost 10000 on Kali and completed the task

